EPFL - Computer Science

COM-407 Advanced Networks

Lecture 05 — January 17, 2026

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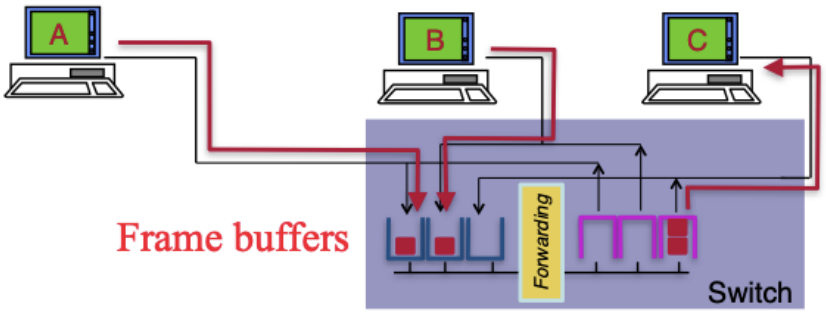
Professor: Pavlos Nikolopoulos

MAC Layer (continued)

2. (Today's) Switched Ethernet

Today, Ethernet is based on:

- switches (or bridges):
  - middle boxes of the MAC layer
  - queing systems (with frame buffers) that forward frames based on MAC addresses
- full-duplex cables:
  - frames flow simultaneously in both directions without suffering collisions (collision-less ethernet)
  - e.g., the well-known UTP cables with separate downlink/uplink

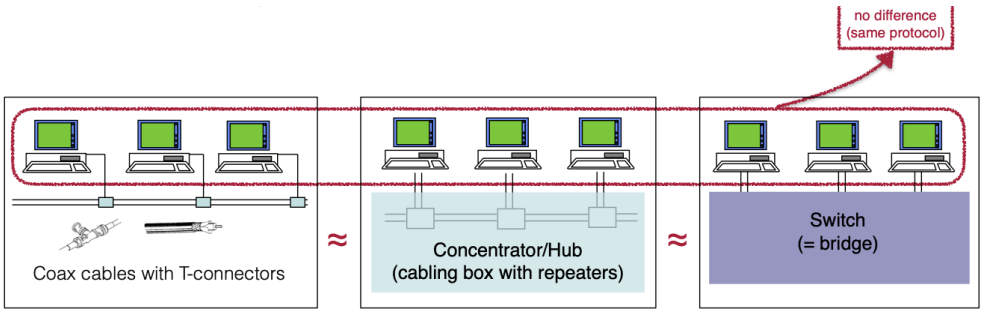


Is switched Ethernet a different protocol?

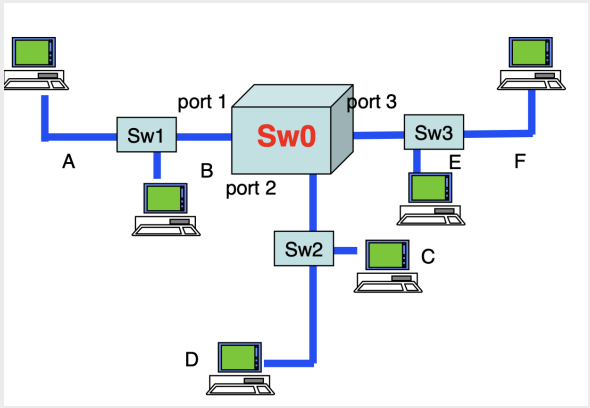
No, switches are designed to be transparent:

i.e. there should be no difference for the protocol used by end-system when they are connected to a coax cable, concentrator/hub or a switch/bridge

But, since we can't get collisions, CSMA/CD is disabled by default.



How many frames can be transmitted in parallel in this switched Ethernet LAN (all boxes are switches)?



A. 1
B. 3
C. 6
D. > 6
E. I don't know

Answer: D

Assume that switches can transmit on all ports in parallel (one LAN adapter per port).

Since switches are full duplex, there can be up to 2 frames transmissions on every link (one in each direction), therefore up to 18 transmissions in parallel

⇒ this is the benefit of today's switches over shared-medium, old-style Ethernet

How does a switch forward a frame?

- Switches use:
  - a forwarding table with exact match
    which maps destination MAC addresses to outgoing interfaces (a.k.a. "ports")

  - and a forwarding algorithm

- Smart switches handle multicast frames differently

Switch's Forwarding Table

Destination MAC address	Outgoing Interface/Port
MAC of A	1
MAC of B	2
MAC of C	3

Switch's Forwarding algorithm

```
Observe all traffic at all ports
For each incoming frame:
  read destination MAC address in MAC header
  if entry for MAC exists in forwarding table:
     obtain matching outgoing port
     if outgoing port = incoming port:
          discard frame (no need to forward)
     else
          forward frame to outgoing port
  else (destination MAC not in the table):
     broadcast to all ports, except for the
     incoming port
```

How does a switch populate its forwarding table?

By observing traffic (plug-and-play device):

- table build by self-learning from source MAC address field in incoming frames
- learnt addresses time out if not re-learnt after an ageing time

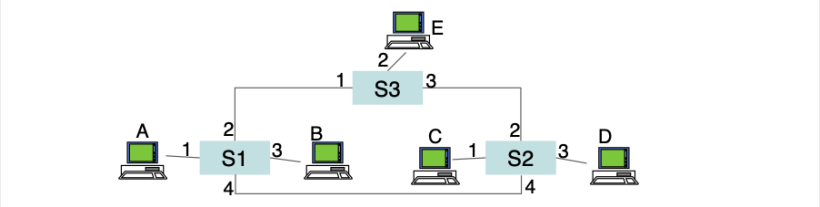
⇒ different from IP routers that use routing algorithms

Can self learning be extended to inter-connected switches?

A. Yes because learning propagates
B. Yes because MAC addresses are unstructured
C. Only in some special cases
D. No, never
E. I don't know

Answer: C

It does not work if there are loops in the topology



E.g.: here S3 does not know whether A is on port 1 or 3, as flooded/broadcast packets may arrive from either side. (Recall that switches are queueing systems, hence there exist non-deterministic delays)

⇒ The entries of the forwarding table may change indefinitely and packets may be trapped in a link, as we saw when talking about TTL in IP layer.

Spanning Tree Protocol (STP): the common way to prevent loops

What does it do?

- forces the active topology to be a tree (i.e. loop-free) that spans all switches by deactivating some links
- adapts to link failures and additions whenever needed

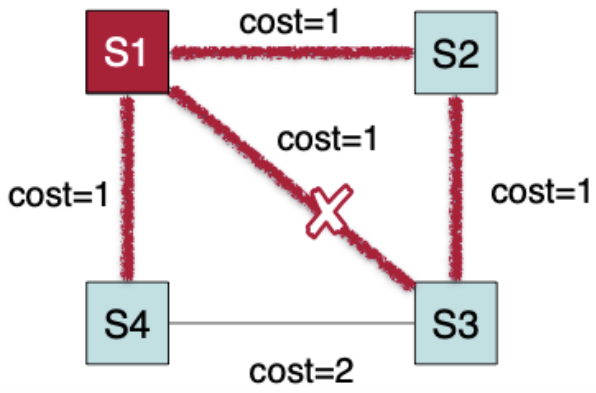
How does the Spanning Tree Protocol work?

The switches:

1. elect a root switch = switch with smallest (configurable) label ID

2. activate only the links that are along the shortest path to root switch. The spanning tree = set of shortest paths to root switch

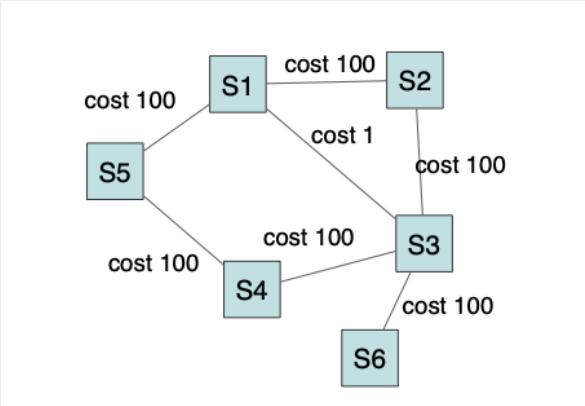
3. monitor whether the root is reachable, and if not, they trigger a re-computation of a new spanning tree



a few important points:

- STP is a distributed algorithm
  - no central control
  - everything is plug and play
  - a gossip protocol is used to learn the IDs
- Shortest paths = paths incurring the smallest aggregate cost to the root
  - each link between switches has a (configurable) cost
  - typically, the cost is a decreasing function of bit rate
- Switches compute the shortest paths using a variant of Bellman-Ford algorithm

Which of the following links are on the spanning tree ?

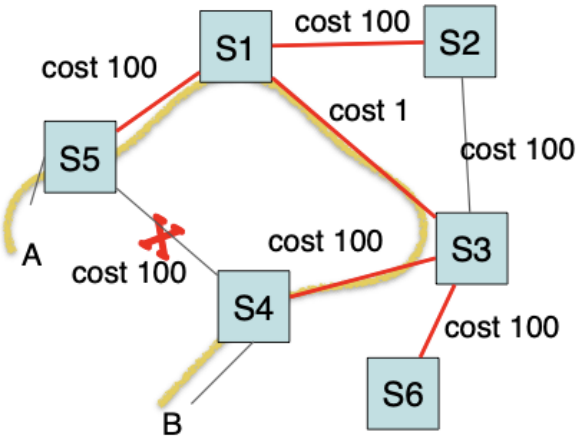


A. S4-S5
B. S4-S3
C. Both
D. None
E. I don't know

Answer: B

A side-effect of the Spanning Tree Protocol (STP)

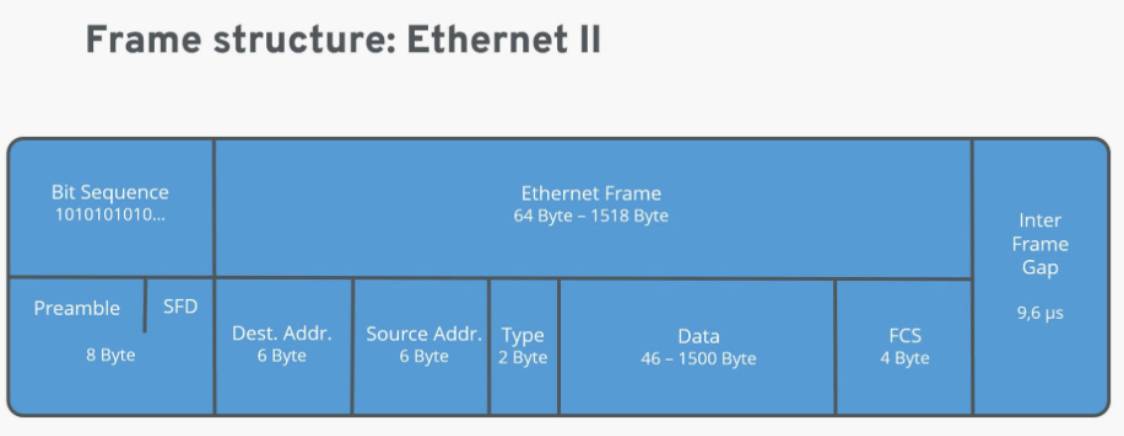
- All frames go through the spanning tree
- A direct link between two (non-root) switches may be not used even if its optimal
  - e.g., the path from A to B, here, is not optimal
- Less efficient than shortest path
- Some more sophisticated switches implement Shortest Path Bridging instead of STP, which is the default



3. Ethernet Frame format

- Ethernet frame = Ethernet's Protocol Data Unit (PDU)
- An Ethernet frame typically carries an IP packet in its payload, sometimes also other:
  - IPv4
  - ARP

- MACSEC (authenticated/encrypted frame)
- Precision Time Protocol



preamble: bits used to detect start of frame necessary to synchronize clocks (e.g., from receiver) which is also called delimiter. It is a bit sequences of 7 bytes, where each byte is equal to: 10101010... and it is terminated by 0. Following the preamble, an additional byte is used to indicate the beginning of the frame. This is called SFD (=start frame delimiter) and its similar to a preamble byte, but its last two bits are 11 (instead of 10).

In Ethernet, transmission starts asynchronously (host transmit independently), and between transmissions, the channel is idle. This is why the clocks of the send and the receiver have to get synchronized at the beginning of each frame.

There is no frame length field in the Ethernet header – its up to the higher layer using Ethernet to find out the length of the payload.

Frames have to be at least 64B (to see why, refer to minimum frame size) and at most 1500B. If needed, padding is inserted to make the frame long enough, and only in that case the Type field is used to indicate the length of the size of the payload before padding.

Error Detection via Frame Check Sequence (FCS)

Why?

- The channel may cause bit errors during transmission ⇒ a frame may be "corrupted"
- We want to detect and discard corrupted frames.

How?

- The sender computes a Cyclic Redundancy Checksum (CRC, 32bits) of the frame and inserts it in the FCS field
- The receiver recomputes the CRC checksum and checks if it is equal to the FCS
- CRC can detect all single, double, triple errors, all error bursts of length ≤ 32, most double burst of length ≤ 17
  - The probability that a random collection of bit errors goes undetected is $\approx 2e - 32$

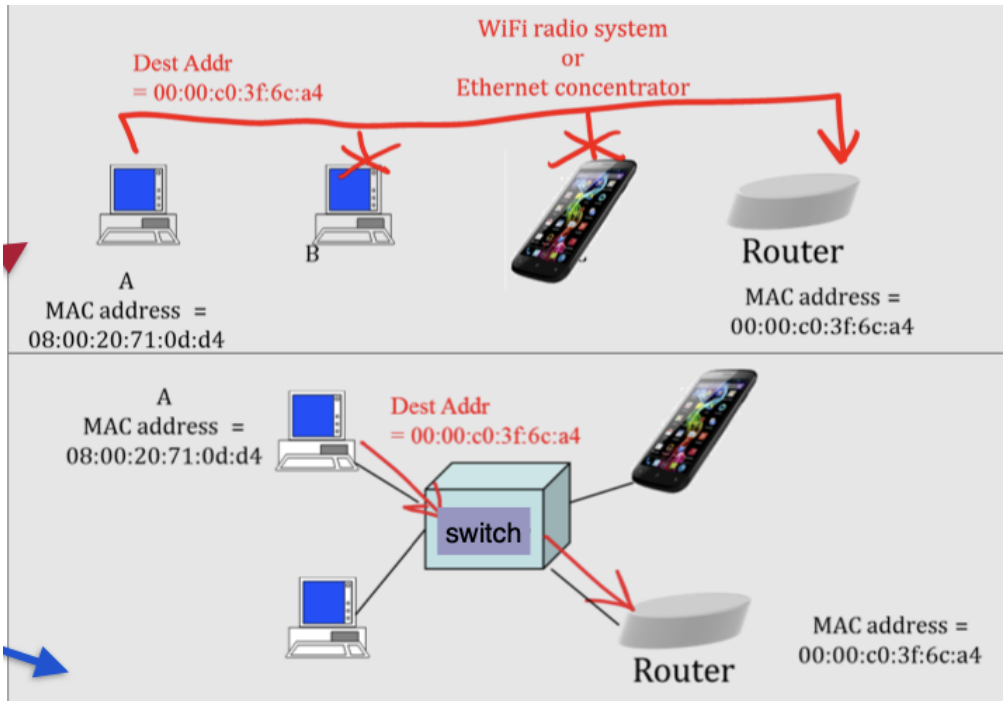
* Error recovery in Ethernet is simple: If a frame fails the check, it is dropped

MAC Unicast

- MAC address: 48 bits = adapted ID, unique worldwide (in principle, but can be also configured sometimes)

How?

- Sender puts destination MAC address in the frame
- Address sent in the cleartext, no encryption
- In shared-medium LAN:
  all stations read all frames, but keep only if the destination address matches
- In switched Ethernet:
  switches forward frames only to ports that need it (in principle, works differently if there's no entry in the forwarding table...)



Addressing: more details (for the interested students)

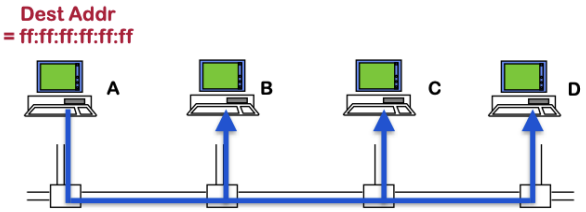
Ethernet addresses are known as MAC addresses. Every Ethernet interface has its own MAC address, which is in fact similar to the serial number of the adapter, set by the manufacturer.

When A sends a data frame to B, A creates a MAC frame with source addr = A, dest addr = B. The frame is sent on the network and recognized by the destination.

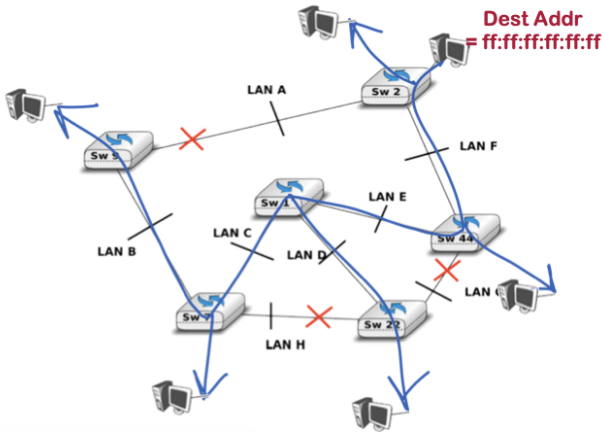
The data is transmitted in Ethernet, with the least significant bit of first octet first – this is a strange behavior dictated by Intel processors. Canonical representation thus inverts the order of bits inside a byte (the first bit of the address is the least significant bit of the first byte)

MAC Broadcast

- ff:ff:ff:ff:ff:ff = broadcast address
- In shared-medium LAN: all machines receive packet and do not discard it



- In switched Ethernet: broadcast frames are sent on all nodes and ports on the spanning tree



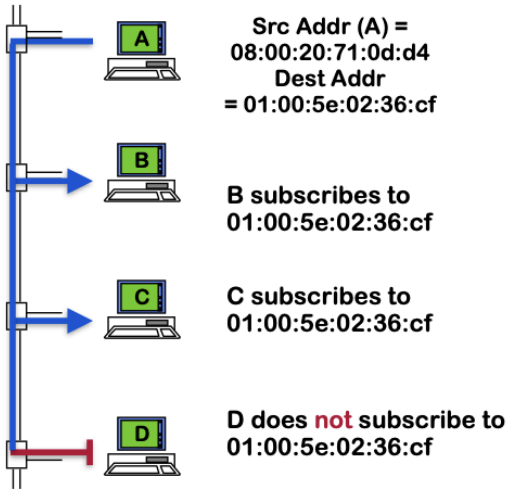
MAC Multicast (= sending to multiple host)

- uses:
  - MAC addresses with 8th bit == 1 are multicast addresses
  - various multicast addresses to support various protocols (e.g., IPv4, IPv6, PTP)

01-00-5e-XX-XX-XX	IPv4 multicast
33-33-XX-XX-XX-XX	IPv6 multicast
01-80-c2-00-01-81	Precision Time Protocol

- makes sense only in switched Ethernet:
  - Ethernet adapter discards multicast packets, unless host subscribes to addresses

- non-smart switches broadcast such frames
- smart switches send only to relevant hosts (the subscribed ones to the multicast group – see "multicast" lecture)



The protocol type in an Ethernet frame header indicates whether...

A. ...the data is a TCP packet or not
B. ...the data is an IPv4 packet or not
C. Both
D. None
E. I don't know

Answer: B

The Ethertype indicates an IPv4 or IPv6 packet

We have a working environment with IPv4, using Ethernet switches plus other things. We add IPv6 to the network. Do we need to change the switch configurations?

A. No, since switches look only at MAC addresses.
B. Yes since the protocol type is different.
C. It depends whether SLAAC is used
D. I don't know

Answer:A

The switches work only at MAC layer and continue to work the same whether IPv4, IPv6 (or something else) is used.

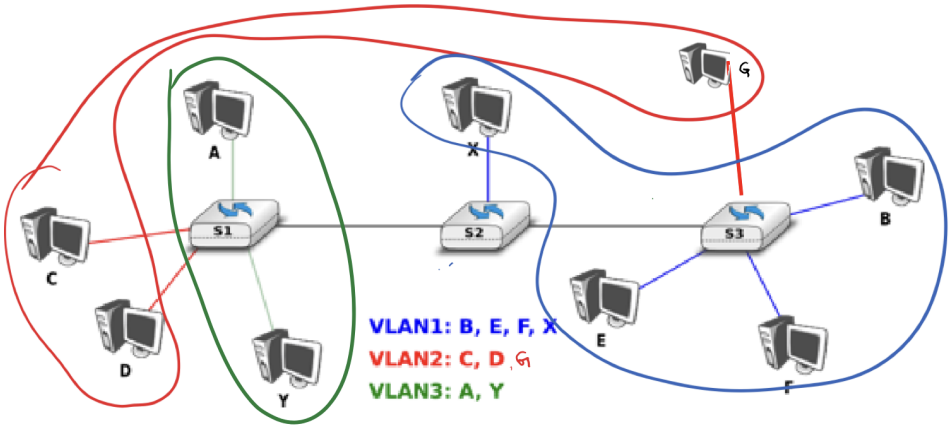
One exception: "smart" switches are cheating – they build multicast trees by looking at IP layer group membership (see multicast lecture later)

4. Virtual LANs

Why invented?

- Lack of traffic isolation → we want to limit the scope of broadcast traffic and have access control
- Inefficient use of switches → we do not want to buy extra switches if other have available ports
- Managing users → if a user moves, we do not want to also change its physical cabling

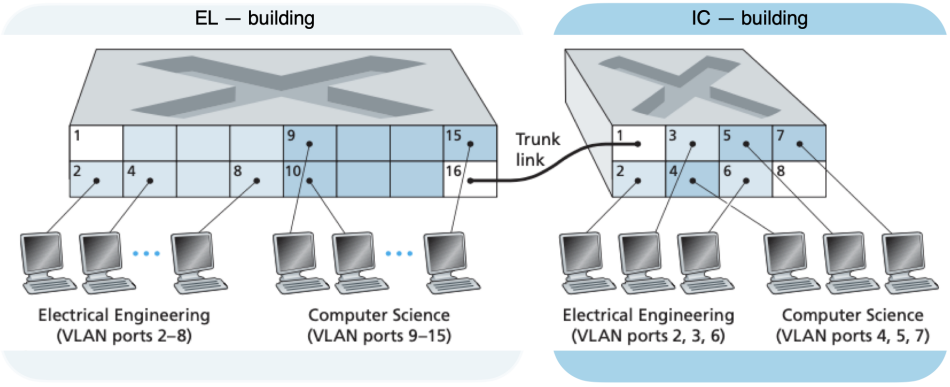
Goal: Decouple who belongs to which LAN from their physical location



- VLANs operate as if they were physically separate:
  - different forwarding tables, different spanning trees
- *How* does it work?

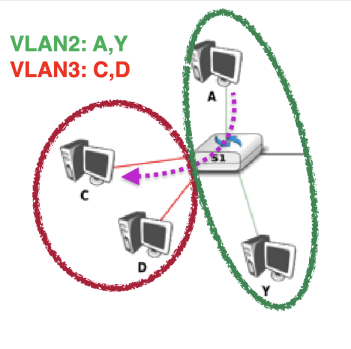
- network manager configures which switch port belongs to which VLAN
- switch treats ports of different VLANs as separate, non communicating LANs
- If more than one switches implement the same VLANs, they are interconnected via *trunk* ports, which use VLAN tags in an Ethernet's *extender header*, in order to correctly associate frames with VLANs.

Example of a trunk port



- The switches implement 2 virtual LANs
  - the Electrical Engineering LAN in light blue and
  - the Computer Science LAN in a blue color
- The switches are interconnected via trunk ports (16 and 1) that belong to both LANS
- The trunk ports use VLAN tags to notify each other which arriving frame belongs to which VLAN

A has an IP packet destined to C; S1 is a switch; what should A do?

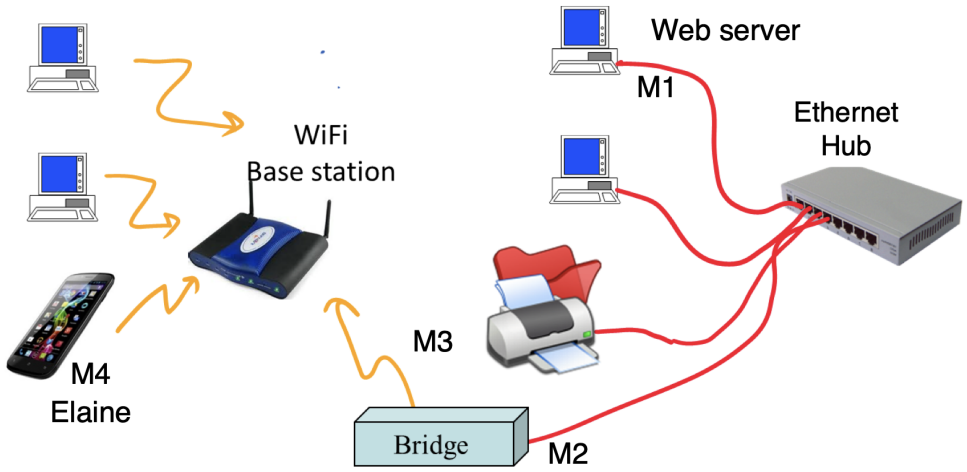


A. Send an ARP request in order to obtain the MAC address of C, then send the packet to that MAC address.
B. Send an ARP request in order to obtain both the MAC address and the VLAN label of C, then send the packet to this MAC address and to this VLAN
C. None of the above
D. I don't know

Answer: C. VLANs 2 and 3 are two *separate* LANs, it is not possible for A to send directly to C - it must go through a router.

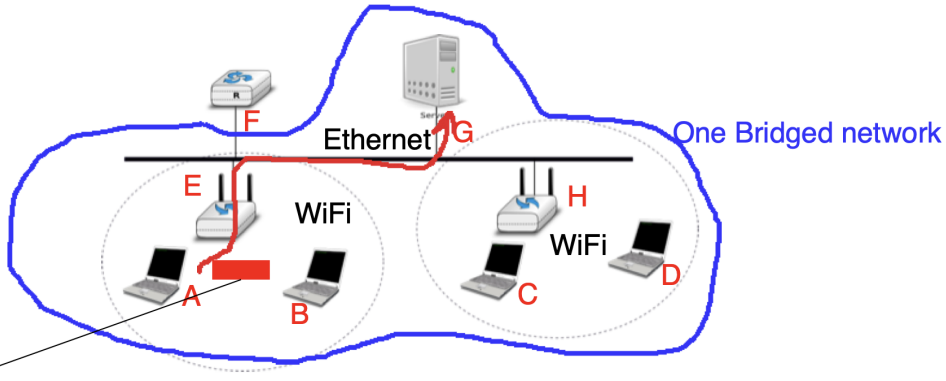
5. Bridges

layer-2 devices that interconnected different medium-access technologies (e.g., WiFi with Ethernet) without routers



WiFi Access Point (a typical bridge)

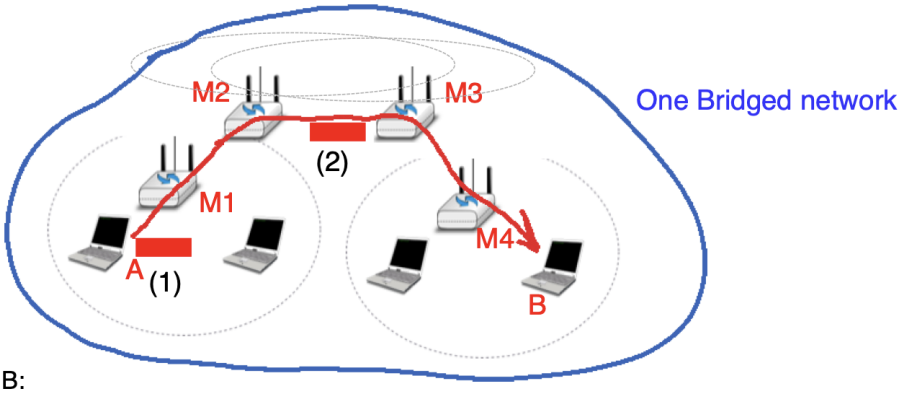
- Combines:
  - a WiFi base station (implements the WiFi MAC protocol)
  - a switch with an Ethernet backend (called the Distribution System)
- WiFi frames differ from Ethernet frames; they include *3 MAC addresses* source, dest and access-point



In frame sent by A to G: source MAC addr = A, dest MAC addr = G, access-point address = E

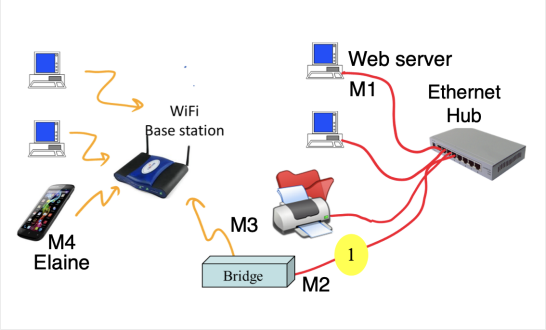
WiFi Access Point with Wireless Distribution System

- WiFi backend instead of Ethernet:
  - WiFi base stations are interconnected via WiFi relays, which act as switches
- WiFi frames include $\leq$ *MAC addresses* source, dest, transmitter/receiver delays



In frame sent by A to B:
At (1) : source MAC addr = A, dest MAC addr = B, access-point address = M1
At (2) : source MAC addr = A, dest MAC addr = B, tx-access-point address = M2, rx-access-point address=M3

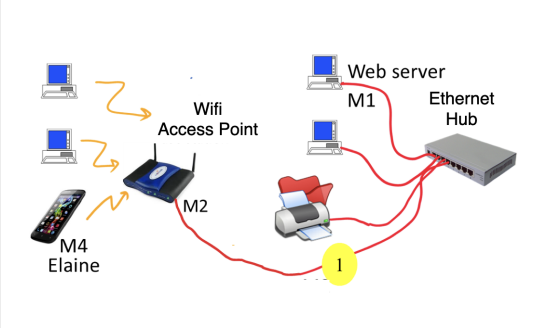
Web server send one IP packet to Elaine. At (1), the MAC destination address is...



A. M1
B. M2
C. M3
D. M4
E. I don't know

Answer: M4

Web server sends one IP packet to Elaine. At (1), the MAC destination address is...



A. M1
B. M2
C. M4
D. I don't know

Answer: M4

Which topology is more efficient? Which one uses resources better?

Answer: the one without the bridge.

In the first case, the WiFi base station is not connected to a wired infrastructure and it acts as a Wifi *range extender*. Instead, it is connected wirelessly to a bridge. This is not very efficient because all WiFi traffic that goes to the bridge uses WiFi twice, once to access the base station, once to go from base station to the bridge.

The second case is the typical use of an *access point*. The base station contains a bridge and is itself connected by Ethernet to the rest of the network.